Geometry
Unit 7
Lesson 3 Practice

Name_Answer Key________
Date____________________
Period__________________

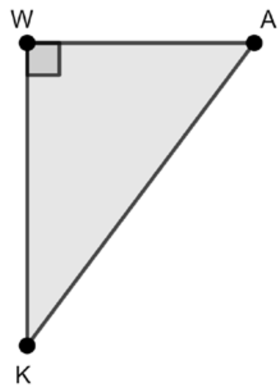
Triangle $WKA$ is shown where $m\angle W = 90°$.

$1.\ 36^2 + b^2 = 85^2$

$1296 + b^2 = 7225$

$\sqrt{b^2} = \sqrt{5929}$

$b = 77$




$2.\ 20^2 + 21^2 = c^2$

$400 + 441 = c^2$

$\sqrt{841} = \sqrt{c^2}$

$c = 29$

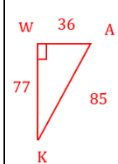
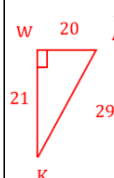
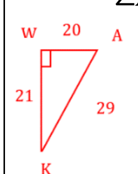
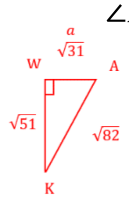
$4.\ a^2 + (\sqrt{51})^2 = (\sqrt{82})^2$

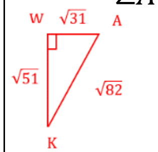
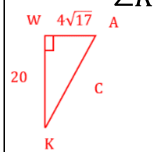
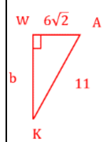
$a^2 + 51 = 82$

$\sqrt{a^2} = \sqrt{31}$

$a = \sqrt{31}$

Complete the table using the given information.

| | $\overline{WA}$ | $\overline{WK}$ | $\overline{KA}$ | **Reference Angle** | **Cosine** | **Sine** | **Tangent** |
|---|---|---|---|---|---|---|---|
| 1. | 36 | 77 | 85 | $\angle A$  | $\frac{A}{H} = \frac{36}{85}$ $\approx .424$ | $\frac{O}{H} = \frac{77}{85}$ $\approx .906$ | $\frac{O}{A} = \frac{77}{36}$ $\approx 2.14$ |
| 2. | 20 | 21 | 29 | $\angle K$  | $\frac{A}{H} = \frac{21}{29}$ | $\frac{O}{H} = \frac{20}{29}$ | $\frac{O}{A} = \frac{20}{21}$ |
| 3. | 20 | 21 | 29 | $\angle A$  | $\frac{A}{H} = \frac{20}{29}$ | $\frac{O}{H} = \frac{21}{29}$ | $\frac{O}{A} = \frac{21}{20}$ |
| 4. | $\sqrt{31}$ | $\sqrt{51}$ | $\sqrt{82}$ | $\angle K$  | $\frac{A}{H} = \frac{\sqrt{51}}{\sqrt{82}}$ | $\frac{O}{H} = \frac{\sqrt{31}}{\sqrt{82}}$ | $\frac{O}{A} = \frac{\sqrt{31}}{\sqrt{51}}$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5. | $\sqrt{31}$ | $\sqrt{51}$ | $\sqrt{82}$ | $\angle A$   | $\frac{A}{H} = \frac{\sqrt{31}}{\sqrt{82}}$ | $\frac{O}{H} = \frac{\sqrt{51}}{\sqrt{82}}$ | $\frac{O}{A} = \frac{\sqrt{51}}{\sqrt{31}}$ |
| 6. | $4\sqrt{17}$ | 20 | $4\sqrt{42}$ | $\angle K$   | $\frac{A}{H} = \frac{20}{4\sqrt{42}}$ | $\frac{O}{H} = \frac{4\sqrt{17}}{4\sqrt{42}} = \frac{\sqrt{17}}{42}$ | $\frac{O}{A} = \frac{4\sqrt{17}}{20} = \frac{\sqrt{17}}{5}$ |
| 7. | $6\sqrt{2}$ | 7 | 11 | $\angle A$   | $\frac{A}{H} = \frac{6\sqrt{2}}{11}$ | $\frac{O}{H} = \frac{7}{11}$ | $\frac{O}{A} = \frac{7}{6\sqrt{2}}$ |
| 8. | 13 | 84 | 85 | $\angle K$ | $\frac{A}{H} = \frac{84}{85}$ | $\frac{O}{H} = \frac{13}{85}$ | $\frac{O}{A} = \frac{13}{84}$ |
| 9. | 8 | 11 | $\sqrt{185}$ | $\angle A$ | $\frac{A}{H} = \frac{8}{\sqrt{185}}$ | $\frac{O}{H} = \frac{11}{\sqrt{185}}$ | $\frac{O}{A} = \frac{11}{8}$ |
| 10. | 9 | 40 | 41 | $\angle K$ | $\frac{A}{H} = \frac{40}{41}$ | $\frac{O}{H} = \frac{9}{41}$ | $\frac{O}{A} = \frac{9}{40}$ |

6. $(4\sqrt{17})^2 + 20^2 = c^2$

$272 + 400 = c^2$

$\sqrt{672} = \sqrt{c^2}$

$\sqrt{16 \cdot 42} = \sqrt{c^2}$

$c = 4\sqrt{42}$

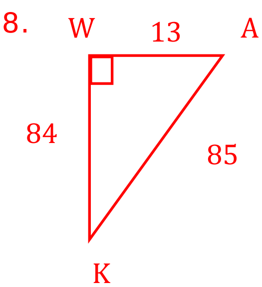
7. $(6\sqrt{2})^2 + b^2 = 11^2$

$72 + b^2 = 121$

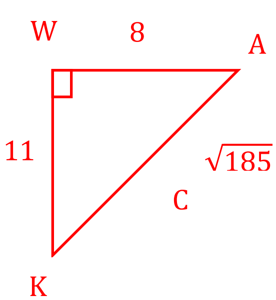
$\sqrt{b^2} = \sqrt{49}$

$b = 7$

8.




9.



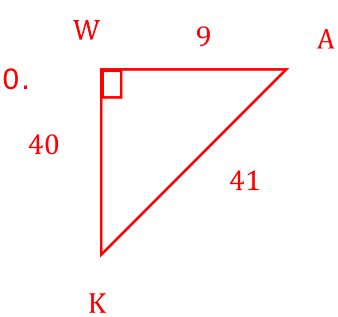

$8^2 + 11^2 = c^2$

$64 + 121 = c^2$

$\sqrt{185} = \sqrt{c^2}$

$c = \sqrt{185}$

10.




$9^2 + 40^2 = c^2$

$81 + 1600 = c^2$

$\sqrt{c^2} = \sqrt{1681}$

$c = 41$